

Flooding in LoRa Mesh Networks

MD Jannatul Rakib Joy

Department of Engineering Technologies
Swinburne University of Technology
Melbourne, Australia
103799644@student.swin.edu.au

Philip Branch

Department of Engineering Technologies
Swinburne University of Technology
Melbourne, Australia
pbranch@swin.edu.au

Abstract—Although LoRa is most well-known for the substantial distances over which it can transmit, it also has some characteristics that make it attractive as a mesh networking technology. In particular, it overcomes one of the main objections to using flooding as a routing mechanism in mesh networking. Flooding involves each node in the network forwarding a copy of every message it receives. Flooding is robust and simple, but unfortunately, if multiple nodes in the mesh receive a message, it typically results in many copies of the same message being received at the destination. However, a LoRa receiver synchronizes with only one transmitter and treats other messages as noise. Because duplicate messages are ignored, flooding can be designed to be much less wasteful of radio resources than mesh networks based on other wireless technologies. In this paper, we demonstrate that LoRa's physical layer mechanism ensures that only one copy of a message is forwarded by a node, regardless of how many copies it receives. We then present results from a simple LoRa mesh network to demonstrate that, for LoRa nodes, flooding is an efficient method of routing.

Index Terms—Mesh networks, Routing, Flooding, LoRa

I. INTRODUCTION

LoRa (Long Range), a low-power, wide-area network (LP-WAN) technology, has emerged as a highly effective solution for IoT applications that demand extensive coverage with minimal power consumption [1]. Operating in unlicensed ISM bands, LoRa is particularly well-suited for sensor and actuator networks due to its low cost, ease of deployment, and impressive Communication range. In optimal conditions, LoRa can achieve communication distances of several kilometres using simple monopole antennas, with even greater ranges possible using directional antennas [2], [3]. This makes LoRa a good choice for a variety of applications, including smart cities, environmental monitoring, and industrial automation.

LoRa is commonly deployed as part of a network architecture known as LoRaWAN, where a gateway mediates communication between LoRa-enabled devices and forwards the data to the Internet using protocols such as MQTT (Message Queuing Telemetry Transport). In a typical LoRaWAN setup, the network is organized in a star topology or a star-of-stars configuration, with LoRa devices communicating directly with a central gateway [4]. However, there is interest in exploring LoRa's potential in mesh network configurations, where communication occurs directly between LoRa-enabled devices without relying on a central gateway. This interest is driven by the need for increased network coverage, redundancy,

and fault tolerance, particularly in remote or challenging environments [5].

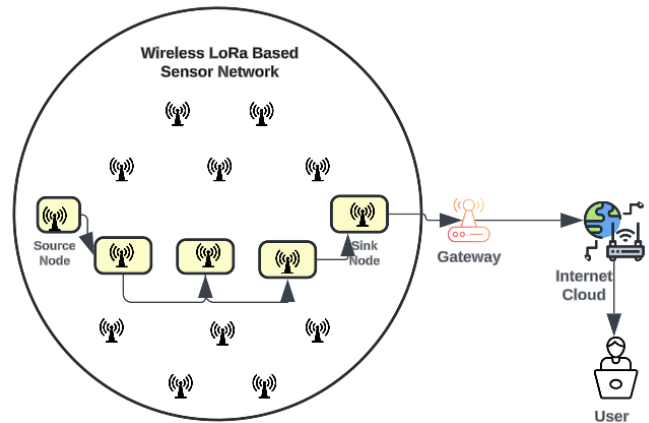


Fig. 1. A simple multi-hop communication in a LoRa mesh network.

Mesh networks, as depicted in Figure 1, are characterized by their multi-hop communication and self-healing capabilities. In a mesh network, data from a source node is relayed through multiple intermediate nodes before reaching the destination node, which then communicates with the gateway and forwards the data to the cloud for processing and storage. This architecture enhances the network's resilience, as it allows the network to continue functioning even if individual nodes fail or become unreachable [6], [7].

However, routing through mesh networks can be challenging. Static routing where a predetermined path is chosen can be very unreliable when a node fails. Routing protocols can consume large amounts of radio resources and power in networks that are commonly resource constrained. Specialised routing protocols have been developed such as OLSR and AODV that are better suited to mesh and ad hoc networks than traditional routing protocols but add to the complexity and hence can be detrimental to the robustness of a mesh network.

A simple approach to routing in a mesh network is simple flooding. In flooding, when a node generates a message, all surrounding nodes within radio range forward the message to their neighbouring nodes. Neighbouring nodes then forward the message to their neighbouring nodes. Eventually the message reaches its destination.

Unfortunately flooding is usually regarded as too wasteful of radio resources to be considered for any other than the most lightly loaded or simplest of networks. One of the attractions of mesh networking is its robustness, but for the mesh to be robust multiple nodes must be in radio range of every other node. Consequently, if flooding is used to route messages, then at each hop in the mesh network many copies of the original message will be forwarded. If each node is in range of k other nodes and there are n hops to the destination, the number of copies of messages at the destination may be of the order k^n .

However, this is not true of LoRa. LoRa is unusual in that a receiver synchronises with a single transmitter for each message. Other messages that are received at the same time are treated as noise. In our previous work with LoRa we made use of this behaviour to deal with contention in linear networks. In this paper we show this behaviour of LoRa has potential for making flooding in LoRa mesh networks an efficient routing mechanism.

The rest of the paper is structured as follows. Section II describes experiments where we demonstrate LoRa only synchronises with a single transmitter even in the presence of multiple transmitters. Section III describes result of the experiments.

II. METHODS

A. Hardware Configuration

We carried out a series of experiments using a mesh topology, where multiple relay nodes facilitated communication between a transmitter (TX) and a receiver (RX). The experiments were structured around eight different scenarios, with each scenario tested three times over a 10-minute duration. Messages were transmitted from the source node at 20-second intervals, which equates to a total of 30 messages sent per scenario. All the tests were conducted at least three times. The nodes were configured to form a simple mesh network, where multiple relay nodes forwarded messages from the transmitter to the receiver via intermediate relay nodes. The LoRa nodes were set up to use 915 MHz, spreading factor of 7 and 125 kHz channels. The number of nodes used in each experiment varied.

B. Network Architectures

The network architectures used for these experiments was simple to complex full mesh network, where messages were relayed from the Transmitter (Tx) through one or more relay nodes before reaching the Receiver (Rx). The goal was to assess how LoRa handles message flooding in this setup, especially focusing on message duplication and the effect of introducing relay delays on overall message delivery.

1) *Single-Hop Multiple Relay*: This section describes the single-hop network architecture used to evaluate LoRa's performance with multiple relay nodes. The goal of this configuration is to explore how LoRa handles message forwarding across a single hop linear network environment with overlapping coverage areas, particularly focusing on message propagation and the role of relays in such a network.

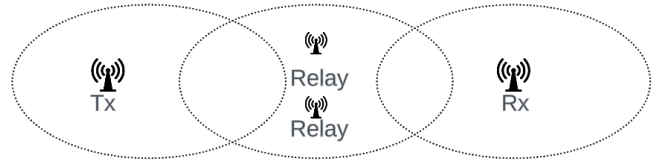


Fig. 2. A single hop Linear network with multiple relays.

In Figure 2, the network configuration includes one transmitter (Tx), two relay nodes, and one receiver (Rx). The transmitter forwards a message, which is relayed by both intermediate nodes before reaching the receiver. The relay nodes are positioned in overlapping coverage areas, ensuring that both relays can receive the same message simultaneously from the transmitter. This design simulates a real-world scenario where multiple relays operate in close proximity, a common setup in IoT networks.

Initially, the network was tested with no delay at the relays. In this no-delay scenario, the relay nodes forwarded messages immediately upon reception. However, this resulted in the receiver discarding duplicate messages due to LoRa's built-in mechanism for handling concurrent transmissions. To mitigate this issue, we introduced a 2-second delay at one of the relay nodes. This delay ensured that the messages from the two relays were forwarded sequentially, allowing the receiver to process both messages without discarding any.

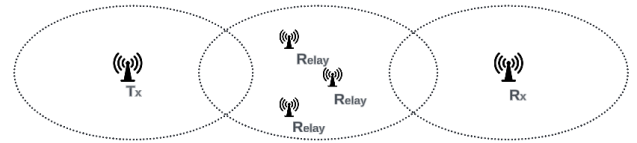


Fig. 3. A single hop Linear network with 3 relays.

In Figure 3, the setup is extended by adding a third relay node, further increasing the complexity of the single-hop configuration. Similar to the two-relay setup, all three relay nodes are positioned in overlapping coverage areas, ensuring simultaneous reception of the message. In the initial no-delay test, all three relays forwarded the same message concurrently, leading to duplicate messages being discarded at the receiver. After the initial test, a 2 & 4 seconds delay was introduced at two of the relays, staggering the message forwarding and preventing duplicate messages from being dropped by the receiver. This approach ensured that the messages reached the receiver in a controlled sequence, reduced collisions, improving the efficiency of message delivery.

2) *Simple to Complex Mesh Networks*: This section focuses on the design of simple to complex mesh networks, where multiple transmitters, receivers, and relay nodes interact. LoRa's unique capability of receiving messages across all nodes within the same coverage area plays a critical role in

these architectures. The addition of controlled delays to the relays is explored to improve message forwarding and reduce the collision and duplication error.

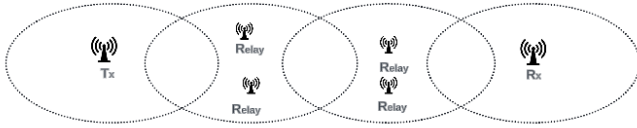


Fig. 4. A simple mesh network with 2 pair of relays .

Figure 4 illustrates a simple mesh network configuration consisting of one transmitter (Tx), one receiver (Rx), and two pairs of relay nodes. In this architecture, the relay nodes are placed in overlapping coverage areas, allowing both relay pairs to receive the transmitted message at the same time. This architecture reflects a common IoT setup where devices rely on multiple transmission paths to ensure message delivery, especially in environments with potential network interference or obstacles. Initially, the network was tested without introducing any delays at the relay nodes. In this no-delay scenario, both relay pairs forwarded the message simultaneously to the receiver. However, due to LoRa's built-in mechanism, the receiver processed only the first message it received and discarded the duplicate as noise. This illustrates how flooding can generate unnecessary packets at the receiver, especially when multiple relay nodes forward the same message at the same time, resulting in redundant copies arriving at the destination.

To illustrate how flooding generates unnecessary packets at the receiver, a 2-second delay was introduced at one of the relay pairs. This delay staggered the transmission of the messages, allowing multiple copies of the same message to arrive at the destination. The receiver, which can only synchronize with one message at a time, now processed more copies of the same message than it would have under normal conditions without delays. This demonstrates that while LoRa synchronizes with one relay, the introduction of relay delays results in many redundant packets, leading to increased flooding and congestion at the receiver.

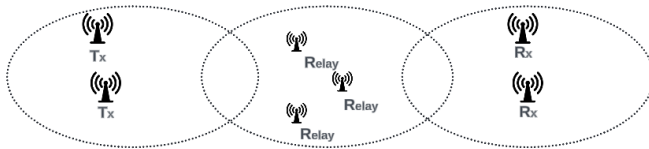


Fig. 5. A full mesh network with 2 different Tx and Rx.

Figure 5 presents a more complex mesh network with two transmitters (Tx), two receivers (Rx), and three shared relay nodes. The goal here was to test how LoRa performs when handling multiple transmissions and shared relay nodes in a mesh configuration. Initially, we conducted the test without any relay delays, but to simulate realistic traffic patterns

and mitigate human error, the second transmitter (TX2) was deliberately delayed by 2 seconds at the start. This staggered start was used to explore how LoRa manages simultaneous transmissions from two transmitters operating with a slight time offset. In this first test, the receivers (Rx1 and Rx2) successfully processed messages from both transmitters, despite the slight time overlap during message forwarding by the relays. LoRa's collision detection mechanism effectively managed the staggered transmissions without significant message loss or duplication. In the second test, a 2-second delay was also introduced at one of the relay nodes to further stagger the message forwarding. While the receivers still synchronized with only one transmission at a time, the staggered relay delays led to an increase in processed messages as the delay caused additional unnecessary copies to reach the receivers, demonstrating how flooding can increase redundant messages. This highlights a key inefficiency in traditional flooding. However, in a LoRa-based network, even when multiple message copies are generated by flooding, LoRa synchronizes with just one message, effectively discarding the redundant copies, which would otherwise overwhelm the receiver in traditional flooding scenarios.

Both Figure 4 and Figure 5 illustrate LoRa's behavior in increasingly complex mesh network configurations, where multiple transmitters and relays share the same communication paths. This demonstrates LoRa's capacity to efficiently manage communication by synchronizing with one transmission at a time, reducing redundancy and ensuring effective message delivery in both simple and complex mesh networks.

III. RESULTS

TABLE I
RESULT OF THE EXPERIMENTS

Message Flooding Results with & without delay		
Test type	Received message	Expected message
Linear network with overlapping of two relays	30	30
Linear network with overlapping of two relays with delays	60	60
Simple Mesh network with overlapping of two sets of two relays	30	30
Simple Mesh network with overlapping of two sets of two relays with delays	Rx-1: 114, Rx 2: 114	120
Linear network with overlapping of three relays	Rx: 30	30
Linear network with overlapping of three relays with delays	Rx: 86	90
Full mesh network with 2 Tx and 2 Rx where one of the Tx was 2 seconds delayed for starting	Rx1: 30 + 30 = 60, Rx2: 30 + 30 = 60	60
Full mesh network with 2 Tx and 2 Rx where one of the Tx was 2 seconds delayed for starting (relay delay)	Rx1: 120, Rx2: 120	120

The experiments in this study evaluated the performance of LoRa-based mesh networks, focusing on the effects of relay delays and message flooding. Table I summarizes the number of successfully received messages compared to the expected count.

The initial experiments focused on linear networks with two and three relays. In the scenario without relay delays, both relays forwarded the messages simultaneously, and the receiver successfully received all 30 messages, as shown in Table I. This demonstrates LoRa's ability to handle message duplication effectively. When a 2-second delay was introduced at the second relay, the network maintained a 100% success rate, with all 60 messages successfully received.

In the configuration with three relays, the network performed similarly. Without delays, all 30 messages were received successfully. After introducing a 2 & 4-seconds-second delay in the second and third relays, 86 out of the expected 90 messages were successfully received. Overall, these results highlight that LoRa's built-in message handling remains effective in linear networks, even as the relay count increases, and timing adjustments play a key role in optimizing performance.

When the network became more complex with multiple relays forming a mesh, the results reflected the challenges of managing overlapping relay coverage areas. Without any delays, one of the receivers missed a single message (29/30), indicating that simultaneous transmissions from different relays could occasionally result in message loss due to collisions. However, LoRa's ability to synchronize with only one message at a time helped minimize the impact of this issue by discarding redundant copies from other relays, even after flooding by both relays at the same time.

In the full mesh network configuration with two transmitters (TX) and two receivers (RX), the initial tests, without relay delays but with a 2-second stagger between transmitters, showed 100% message delivery (60/60 messages), as seen in Table I. This demonstrates LoRa's ability to handle simultaneous transmissions effectively when slightly offset.

When a 2-second delay was introduced at one of the transmitters and later at one of the relays, Each receiver logged 120 received messages, even though only 60 were originally sent. Delays allowed for redundant message forwarding, minimizing collisions and creating numerous duplicate messages. However, in scenarios without delay, LoRa's built-in synchronization mechanism ensures that only one message is processed at a time, even in flooding scenarios. This shows LoRa's ability to handle flooding more efficiently than traditional networks, where without delay, it discards redundant copies by synchronizing with just one message at a time, minimizing unnecessary duplications that would typically overwhelm a network.

These findings highlight that LoRa's broadcasting behavior, significantly enhances message reliability in complex networks. This makes LoRa a promising solution for IoT mesh networks, where overlapping transmission paths and multiple relays often pose challenges. Additionally, unlike

other technologies, LoRa synchronizes with all nearby relays, ensuring that it receives messages even when multiple relays transmit simultaneously. This unique capability enables LoRa to efficiently manage message flooding, avoiding duplication or loss, and reinforcing its effectiveness in complex mesh network scenarios.

The results demonstrate that LoRa's built-in synchronization ensures that only one message is processed during flooding, effectively minimizing unnecessary duplicates and enhancing message reliability, making it an ideal solution for complex IoT mesh networks with overlapping relays.

IV. CONCLUSION

In this paper we have conducted experiments that demonstrate's LoRa receiver behaviour when it receives multiple messages from multiple transmitters make it an excellent technology for mesh networking. Unlike other technologies it synchronises with just one of the messages from nearby transmitters and treats the other messages as noise. This has the highly desirable outcome that when such nodes are meshed into a multi-hop mesh network, and flooding is used for routing message, only a single copy of the message is transmitted beyond each hop thus avoiding the wastefulness usually associated with flooding.

Future research will involve exploring the capacity and scalability of such networks. How many nodes can such a network support? It will explore the potential of using it for mobile ad hoc networks where the nodes are in motion relative to each other.

We believe LoRa has great potential as a mesh network technology. We hope our future research will realise some of that potential.

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